

Education

PhD	Stanford University	Aeronautical and Astronautical Engineering	2020
	<i>Co-advisors:</i>	<i>Gianluca Iaccarino and Art B. Owen</i>	
MS	Stanford University	Aeronautical and Astronautical Engineering	2018
BS	Olin College of Engineering	Mechanical Engineering	2014

PhD Thesis

“Precision Margin: First-Principles Margins for Aircraft Design Under Uncertainty”

Competitive aircraft design walks a fine line, balancing weight reduction with aircraft safety. Design tends to lean conservative, with cascaded margins to address uncertainties. This thesis introduces design margins that *provably* yield minimal weight penalties at desired levels of safety. Comparisons against industry standards, tractable approximations, and ramifications for both conceptual and detailed design are considered.

Grants and Fellowships

Source	State	Size	Start
Towards Data-Capable Engineers with a Variability-Capable Mindset <i>NSF EEC Award Number 2138463</i>	Awarded	\$148,189	2022
Building and Sustaining a Diverse Academic Community <i>Stanford Alumni Association</i>	Complete	\$5,370	2021
Improving User Understanding of the Catalyst Game <i>Grant from Toyota Research Institute to fund usability study</i>	Complete	~\$20,000	2021
Integrated Project Fund <i>Grant to develop TA training workshop, Olin internal</i>	Complete	~\$2,530	2021
Diversifying Academia, Recruiting Excellence (DARE) Fellowship <i>Competitive fellowship for promising faculty candidates, Stanford internal</i>	Complete	~\$116,000	2018
NSF Graduate Research Fellowship	Complete	~\$300,000	2015

Honors and Awards

Patricia Cross Future Leaders Award <i>Competitive award for graduate students with promise as future leaders of higher education; administered by the Association of American Colleges & Universities</i>	2020
Stanford MECON Oral Presentation award, 1st place <i>Mechanical Engineering Department-sponsored speaker competition</i>	2017
AIAA Jefferson Goblet Best Student Paper <i>Highest honor for student papers at AIAA SciTech annual conference</i>	2017

Books

1. **del Rosario, Z.** and G. Iaccarino (2024). *Computational Modeling by Case Study: All Models are Uncertain*. Cambridge Scholars Publishing. eprint: <https://www.cambridgescholars.com/product/978-1-0364-0291-4>.

Journal Articles (peer reviewed)

1. **del Rosario, Z.** (2024a). Neglected, Acknowledged, or Targeted: A Conceptual Framing of Variability, Data Analysis, and Domain Consequences. *Journal of Statistics and Data Science Education*. eprint: <https://doi.org/10.1080/26939169.2024.2308119>.

2. Hegde, V. I., C. K. H. Borg, **Z. del Rosario**, Y. Kim, M. Hutchinson, E. Antono, J. Ling, P. Saxe, J. E. Saal, and B. Meredig (2023). Quantifying uncertainty in high-throughput density functional theory: A comparison of AFLOW, Materials Project, and OQMD. *Phys. Rev. Mater.* 7 (5), 053805. eprint: <https://link.aps.org/doi/10.1103/PhysRevMaterials.7.053805>.

3. **del Rosario, Z.** (2022a). An Open-Source Active Learning Curriculum for Data Science in Engineering. *Journal of Open Source Education* 5(49), 117. eprint: <https://doi.org/10.21105/jose.00117>.
4. **del Rosario, Z.** (2021a). Precision Materials Indices: Materials Selection with Statistically-rigorous Reliability Analysis. *AIAA Journal*. eprint: <https://arc.aiaa.org/doi/full/10.2514/1.J060521>.
5. **del Rosario, Z.**, G. Iaccarino, and R. W. Fenrich (2021). When Are Allowables Conservative? *AIAA Journal*. eprint: <https://arc.aiaa.org/doi/full/10.2514/1.J059578>.
6. **del Rosario, Z.** (2020c). Grama: A Grammar of Model Analysis. *Journal of Open Source Software* 5(51), 2462. eprint: <https://doi.org/10.21105/joss.02462>.
7. **del Rosario, Z.**, M. Rupp, Y. Kim, E. Antono, and J. Ling (2020). Assessing the frontier: Active learning, model accuracy, and multi-objective candidate discovery and optimization. *The Journal of Chemical Physics* 153(2), 024112. eprint: <https://aip.scitation.org/doi/pdf/10.1063/5.0006124>.
8. Jofre-Cruanyes, L., **Z. del Rosario**, and G. Iaccarino (2020). Data-driven dimensional analysis of heat transfer in irradiated particle-laden turbulent flow. *International Journal of Multiphase Flow*. eprint: <https://doi.org/10.1016/j.ijmultiphaseflow.2019.103198>.
9. **del Rosario, Z.**, R. Fenrich, and G. Iaccarino (2019a). Cutting the Double Loop: Theory and Algorithms for Reliability-Based Design Optimization with Parametric Uncertainty. *International Journal for Numerical Methods in Engineering*. eprint: <https://doi.org/10.1002/nme.6035>.
10. **del Rosario, Z.**, G. Iaccarino, and R. W. Fenrich (2019). Fast Precision Margin with the First-Order Reliability Method. *AIAA Journal*. eprint: <https://doi.org/10.2514/1.J058345>.
11. **del Rosario, Z.**, M. Lee, and G. Iaccarino (2019). Lurking Variable Detection via Dimensional Analysis. *SIAM / ASA Journal on Uncertainty Quantification*. eprint: <https://doi.org/10.1137/17M1155508>.

Conference Papers (peer reviewed)

[†] indicates student co-authors

1. Bostwick[†], L. and **Z. del Rosario** (2026). GIFTS: An Activity to Enhance Engineering Modeling with an Entrepreneurial Mindset. In: *ASEE Annual Conference and Exposition*.
2. Boyd[†], E., L. Fries[†], S. Jalinous[†], and **Z. del Rosario** (2026). Tenchi Diagrams: A Technique to Make Visible Student Thinking About Models. In: *12th International Conference on Teaching Statistics*.
3. **del Rosario, Z.** and G. Van de Zande (2026). EEC, ERI: A Mixed-Methods Study of Decision-making Under Variability. In: *ASEE Annual Conference and Exposition*.
4. George[†], A., L. Bostwick[†], and **Z. del Rosario** (2026). Being vs. Doing: Student Insights on Engineering Identity from a Collaborative AutoEthnography. In: *ASEE Annual Conference and Exposition*.
5. **del Rosario, Z.** (2025b). ERI: A Mixed-Methods Study of Statistical Thinking, Informed by Engineering Practice. In: *ASEE Annual Conference and Exposition*.
6. George[†], A., L. Bostwick[†], A. Renaud[†], E. Boyd[†], and **Z. del Rosario** (2025). Research as Teaching: On Student Mindset and Voice in a Sustained Collaborative AutoEthnography on Mathematical Modeling. In: *ASEE Annual Conference and Exposition*. eprint: <https://peer.asee.org/57130>.
7. Matsumoto, S., **Z. del Rosario**, J. Huang, C. Lynch, K. Mbanisi, P. Ruvolo, and G. van de Zande (2025). Lessons Learned from Helping Faculty "Make the Pivot" to New Research Areas through a Community of Practice. In: *ASEE Annual Conference and Exposition*.

8. Abbot[†], E., B. Lan[†], L. Raus[†], B. Fan[†], and **Z. del Rosario** (2024). Beyond "How's it going?": A collaborative Autoethnographic Study by Early Instructors in a First-Year Engineering Studio Course. In: *ASEE Annual Conference and Exposition*. eprint: <https://nemo.asee.org/public/conferences/344/papers/41115/view>.
9. Bostwick[†], L., A. George[†], T. Lee[†], and **Z. del Rosario** (2024). When Is It Relevant? A Collaborative Autoethnography Study by Engineering Students on Statistical Variability. In: *ASEE Annual Conference and Exposition*. eprint: <https://nemo.asee.org/public/conferences/344/papers/41422/view>.
10. **del Rosario, Z.**, J. Ryu[†], and E. Saur[†] (2024a). A Mixed-Methods Study of Statistical Thinking in Engineering Practice. In: *ASEE Annual Conference and Exposition*. eprint: <https://nemo.asee.org/public/conferences/344/papers/41041/view>.
11. **del Rosario, Z.**, J. Ryu[†], and E. Saur[†] (2024b). Targeting Consequences of Variability as a Data Science Resource. In: Auckland, NZ: International Association for Statistical Education. eprint: <https://doi.org/10.52041/iase24.505>.
12. Fox[†], E. and **Z. del Rosario** (2024). A Qualitative Study of Engineers' Perception of Variability as 'Error'. In: *ASEE Annual Conference and Exposition*. eprint: <https://nemo.asee.org/public/conferences/344/papers/40999/view>.
13. **del Rosario, Z.** (2023a). A Mixed-Methods Investigation of Engineers Targeting the Consequences of Variability. In: *ASEE Annual Conference and Exposition*. eprint: <https://nemo.asee.org/public/conferences/327/papers/36726/view>.
14. Raus[†], L., K. Mackowiak[†], S. Matsumoto, and **Z. del Rosario** (2023). The Meme Game: A Hands-On Activity to Introduce First-Year Engineers to Concepts in Mathematical/Computational Modeling. In: *ASEE Annual Conference and Exposition*. eprint: <https://nemo.asee.org/public/conferences/327/papers/39649/view>.
15. Vo[†], K., A. Evans[†], S. Madan[†], and **Z. del Rosario** (2023). A Scoping Review of Engineering Textbooks to Quantify the Teaching of Uncertainty. In: *ASEE Annual Conference and Exposition*. eprint: <https://nemo.asee.org/public/conferences/327/papers/36798/view>.
16. **del Rosario, Z.** (2022). Enabling Active Learning Pedagogy and Insight Mining with a Grammar of Model Analysis. In: *Proceedings of the 21st Python in Science Conference*. Ed. by M. Agarwal, C. Calloway, D. Niederhut, and D. Shupe, pp.209–216. eprint: https://conference.scipy.org/proceedings/scipy2022/pdfs/zachary_delrosario.pdf.
17. Aggarwal[†], R., M. Flynn[†], S. Daitzman[†], D. Lam, and **Z. del Rosario** (2021). A Qualitative Study of Engineering Students' Reasoning About Statistical Variability. In: *ASEE Middle Atlantic Section, Conference Proceedings*.
18. **del Rosario, Z.**, R. Aggarwal[†], C. Coffey[†], A. Sadler[†], S. Matsumoto, A. Wood, P. Ruvolo, and J. Woodard (2021). Work in Progress: Crafting a Virtual Studio: Some Models and Implementations. In: *ASEE Annual Conference and Exposition, Conference Proceedings*.
19. **del Rosario, Z.**, R. W. Fenrich, and G. Iaccarino (2020). When are Design Allowables Conservative? In: *AIAA SciTech 2020 Forum*.
20. **del Rosario, Z.**, R. W. Fenrich, and G. Iaccarino (2019a). Beyond Basis Values: Fast Precision Margin with FORM. In: *21st AIAA Non-Deterministic Approaches Conference*.
21. **del Rosario, Z.**, R. W. Fenrich, and G. Iaccarino (2019b). Margin as Model: Some Answers to "How Many Tests Should I Perform?". In: *AIAA Aviation 2019 Forum*.
22. **del Rosario, Z.**, A. Towne, and G. Iaccarino (2018b). Dimension Reduction for Shape Design Insight. In: *20th AIAA Non-Deterministic Approaches Conference*.
23. **del Rosario, Z.**, P. Constantine, and G. Iaccarino (2017c). Developing Design Insight Through Active Subspaces. In: *19th AIAA Non-Deterministic Approaches Conference*.

Preprints

[†] indicates student co-authors

1. **del Rosario, Z.**, E. Saur[†], J. Ryu[†], J. Lin[†], and J. B. Wilmer (2026). DECISIONS, VARIABILITY, AND VISUALIZATION: A NOVEL INSTRUMENT FOR DECISION MAKING UNDER VARIABILITY. *PsyArxiv*. eprint: <https://doi.org/10.31234/osf.io/vujta.v3>.
2. Fox[†], E., A. Evans[†], K. Vo[†], E. Ramos[†], M. Stites[†], and **Z. del Rosario** (2022). Context-Engaged Engineering Data Analysis: A Grounded Theory of Engineers Targeting the Consequences of Variability. *PsyArxiv*. eprint: <https://psyarxiv.com/q6a3j/>.
3. Jofre-Cruanyes, L., **Z. del Rosario**, and G. Iaccarino (2019). Dimension reduction of thermo-fluid mechanisms in irradiated particle-laden turbulence. *Center for Turbulence Research Annual Research Briefs*.
4. Constantine, P. G., **Z. del Rosario**, and G. Iaccarino (2017). Data-driven dimensional analysis: algorithms for unique and relevant dimensionless groups. *arXiv preprint arXiv:1708.04303*. Forthcoming in JCP.
5. Constantine, P. G., **Z. del Rosario**, and G. Iaccarino (2016). Many physical laws are ridge functions. *arXiv preprint arXiv:1605.07974*.

Invited Talks

1. **del Rosario, Z.** (2025e). Targeting Variability: Drawing Connections Between Data Science and Engineering Practice to Inform Teaching. In: Engineering Education Transformations Institute, UGA. <https://www.youtube.com/watch?v=Oq54dQns2ow>.
2. **del Rosario, Z.** (Oct. 2023b). Targeting Variability: Combining Statistical Thinking and Engineering Practice. In: Tufts, ME Seminar.
3. **del Rosario, Z.** (Aug. 2023c). Targeting Variability: Ideas for addressing the consequences of variability. In: SpaceX, Build Reliability Engineering.
4. **del Rosario, Z.** (Apr. 2021c). Studio Teaching Online: Lessons learned from a mixed-methods study. In: Stanford University.
5. **del Rosario, Z.** (Nov. 2020a). Assessing the Frontier. In: Cambridge Cheminformatics Meeting. <http://www.c-inf.net/>.
6. **del Rosario, Z.** (Feb. 2020d). Grammar and Margins. In: Toyota Research Institute.
7. **del Rosario, Z.** (Oct. 2019a). Aircraft Design Under Uncertainty. In: Harvey Mudd College Seminar.
8. **del Rosario, Z.** (Oct. 2019d). The Curse of Dimensionality: Problems and Strategies. In: NATO/STO Lecture Series: Uncertainty Quantification in Computational Fluid Dynamics. <https://we.stanford.edu/LSUQ>.
9. **del Rosario, Z.**, R. Fenrich, and G. Iaccarino (July 2019b). Principled Margin. In: Arevo, Inc.
10. **del Rosario, Z.** (Sept. 2018a). Lost in Hyperspace: The Curse of Dimensionality. In: Wellesley College student seminar.
11. **del Rosario, Z.** (Oct. 2018b). The Curse of Dimensionality: Problems and Strategies. In: von Karman Institute: Uncertainty Quantification in Computational Fluid Dynamics (STO-AVT 326).
12. **del Rosario, Z.**, A. Towne, and G. Iaccarino (2018c). Dimension Reduction for Shape Design Insight. In: Aerospace Computational Design Lab (ACDL) seminar, MIT.

Conference Talks

1. **del Rosario, Z.** (2022b). Beyond Aleatory—A Taxonomy of Uncertainty for Reduction and Consequence. In: Stanford University. <https://icme.stanford.edu/events/seminar/beyond-aleatory-taxonomy-uncertainty-reduction-and-consequence>.

2. **del Rosario, Z.** (2022c). Coding to Learn: Reframing Programming in Teaching. In: *BOW Forum on Innovative Teaching*.
3. **del Rosario, Z.**, L. Jofre, and G. Iaccarino (2022). Uncovering the link between dimensional analysis and causality. In: *Causality in Turbulence and Transition*. Universidad Politécnica Madrid. eprint: <https://torroja.dmt.upm.es/causturb/static/assets/abstracts/Iaccarino.pdf>.
4. **del Rosario, Z.** and G. Iaccarino (2020). Physics-informed Inference: Dimensional Analysis as Dimension Reduction. In: SIAM UQ (Conference Cancelled).
5. **del Rosario, Z.** (2019b). Machine Learning for Materials Property Prediction. In: North American Solid State Chemistry Conference.
6. **del Rosario, Z.** (2019c). Stanford SeeME: Student-driven research within an R1 institution. In: National Alliance for Broader Impacts (NABI) Summit.
7. **del Rosario, Z.**, A. Banko, A. Horwitz, and G. Iaccarino (2018). Data-Driven Physical Inquiry: Discovering Relevant Dimensionless Numbers With Physics-Constrained Machine Learning. In: 71th Annual Meeting of the American Physical Society, Division of Fluid Dynamics.
8. **del Rosario, Z.**, A. Towne, and G. Iaccarino (2018a). Dimension Reduction for Shape Design Insight. In: Thermal, Fluid science Sponsors, and Affiliates conference (TFSA).
9. **del Rosario, Z.**, P. Constantine, and G. Iaccarino (2017a). Algorithm-Driven Insight. In: Thermal and Fluid Science Affiliates Conference.
10. **del Rosario, Z.**, P. Constantine, and G. Iaccarino (2017b). Data-Driven Dimensional Analysis. In: CompFest.
11. **del Rosario, Z.** and G. Iaccarino (2017). Hidden Parameter Hypothesis Testing. In: Statistical Perspectives on Uncertainty Quantification.
12. **del Rosario, Z.**, M. Lee, and G. Iaccarino (2017). Discovering Hidden Controlling Parameters using Data Analytics and Dimensional Analysis. In: 70th Annual Meeting of the American Physical Society, Division of Fluid Dynamics.
13. **del Rosario, Z.**, A. Towne, and G. Iaccarino (2017). Handling Classes of Variables in Dimension Reduction. In: SIAM Workshop on Parameter Space Dimension Reduction (DR17).
14. Torres, H., **Z. del Rosario**, and G. Iaccarino (2017). MCRT. In: WEST Conference.
15. **del Rosario, Z.** and G. Iaccarino (2016). Pi Active Subspace. In: Thermal and Fluid Science Affiliates Conference.

Panel Appearances

1. **del Rosario, Z.** (Jan. 2022d). Professors for the Future. In: UC Davis.
2. **del Rosario, Z.** (May 2021b). Scientific Teaching Institute Alumni Panel. In: SEPAL Lab.
3. **del Rosario, Z.** (Aug. 2020g). Project Based Learning at the College/University Level. In: Strive Virtual College Exploration STEM Days, Cachet / StriveScan. <https://www.strivescan.com/virtual/stem/>.

Poster Presentations

1. **del Rosario, Z.** (2025c). Measuring Targeting: Development and Deployment of a Novel Instrument for Decision-Making Under Variability. In: US Conference on Teaching Statistics (USCOTS), Research Satellite.
2. **del Rosario, Z.** (2025d). Surprise—they're different! An introductory statistics activity to highlight differences between Frequentist and Bayesian statistics. In: US Conference on Teaching Statistics (USCOTS).

Other Articles

1. **del Rosario, Z.** (2025a). All models are wrong – a computational modeling expert explains how engineers make them useful. *The Conversation*. eprint: <https://theconversation.com/all-models-are-wrong-a-computational-modeling-expert-explains-how-engineers-make-them-useful-253309>.
2. Langehennig, S., **Z. del Rosario**, and M. Dogucu (2025). Surprise, They're Different! Comparing Frequentist and Bayesian Instructional Approaches in Political Science and Public Policy Classrooms. *Political Science Educator* 29 (1). eprint: <https://educate.apsanet.org/surprise-theyre-different-comparing-frequentist-and-bayesian-instructional-approaches-in-political-science-and-public-policy-classrooms>.
3. **del Rosario, Z.** (2024b). The 'average' revolutionized scientific research, but overreliance on it has led to discrimination and injury. *The Conversation*. eprint: <https://theconversation.com/the-average-revolutionized-scientific-research-but-overreliance-on-it-has-led-to-discrimination-and-injury-223509>.
4. **del Rosario, Z.** and M. del Rosario (2022). Synthesizing domain science with machine learning. *Nature Computational Science*. eprint: <https://www.nature.com/articles/s43588-022-00358-2>.
5. **del Rosario, Z.** (2021d). We're Not Going to Build It: Why STEM students need to learn design refusal. *Liberal Education*. eprint: <https://www.aacu.org/liberaleducation/articles/why-stem-students-need-to-learn-design-refusal>.
6. **del Rosario, Z.** (2020b). Closing the Gap: Perspectives from a Cross Scholar on Advancing Diversity, Equity, and Inclusion. *AAC&U News, Perspectives*. eprint: <https://www.aacu.org/aacu-news/newsletter/closing-gap-perspectives-cross-scholar-advancing-diversity-equity-and-inclusion>.
7. **del Rosario, Z.** (2020e). Olin "Faux-mencement": A Case Study in Cocreation. *AAC&U Blog*. eprint: <https://www.aacu.org/liberaleducation/articles/olin-faux-mencement-a-case-study-in-cocreation>.
8. Yegnashankaran, K. and **Z. del Rosario** (2020). 10 Strategies for Collegial Videoconferencing. *Stanford Center for Teaching and Learning*. eprint: <https://docs.google.com/document/d/1raLzjieDcVtvLU6-6wi5rS81Odm0UWLG1UeH2RECdDo/edit#heading=h.ej4ozbwqqyvk>.

Reviewing

I review manuscripts for the following journals:

- [ASA Journal of Statistics and Data Science Education](#): 5x
- [American Journal of STEM Education](#): 2x
- [Statistics Education Research Journal](#): 1x
- [American Institute of Aeronautics and Astronautics Journal](#): 2x
- [Nature Computational Science](#): 1x
- [Data-Centric Engineering](#): 1x
- [Progress in Materials Science](#): 1x
- [Physics of Fluids](#): 1x

Teaching Experience

Courses Taught	Offered (Last Taught)
Modeling and Simulation of the Physical World (MTH 1111) Olin <i>A studio-based class on scientific modeling: building models at multiple levels of abstraction, tailoring them for scientific questions, implementing models in software. Students also practice working in teams and communicating scientific conclusions.</i>	Fall (2025)
Quantitative Engineering Analysis 3 (ENGRX 2011) Olin <i>A studio-based course on differential equations and fourier analysis.</i>	Fall (2025)
Data Science (MTH 2131) Olin <i>A flipped, studio-based class on data fundamentals: extract-transform-load operations, exploratory data analysis (EDA), inferential statistics, regression analysis. https://github.com/zdelrosario/data-science-curriculum</i>	Spring (2024)
Collaborative Design (ENGR 3299) Olin <i>A studio-based introduction to the human-centered design process: need finding, persona creation, idea development and shaping.</i>	Spring (2022)
Quantitative Engineering Analysis 2 (ENGR 2005) Olin <i>A studio-based course on engineering analysis fundamentals: multivariable calculus, statics, optimization.</i>	Spring (2022)
Probabilistic Design Optimization (ENGR 3299) Olin <i>Fundamentals of optimization and probability for design: multi-objective, constrained optimization, copula probability modeling, Monte Carlo and FORM approximation.</i>	Spring (2021)
Summer Data Science - Olin <i>A five-week summer course for Olin alumni. Designed open-source self-paced curriculum, reinforced with team-based data challenges. Created remote learning environment around team cohorts using Discord. https://github.com/zdelrosario/data-science-curriculum</i>	Summer (2020)
Uncertainty Quantification (ME 470) Stanford <i>A graduate-level elective course for 9 advanced students. Taught using a mixture of lecture and evidence-based methods.</i>	Spring (2019)

Teaching Competencies

Engineering	Fluid Dynamics, Solid Mechanics, Aerodynamics, Human-centered Design
Statistics	Probability, Estimation, Design of (Computer) Experiments, Regression, EDA
Mathematics	Linear Algebra, Ordinary/Partial Differential Equations, Optimization
Computing	Python (Numpy/Scipy, Pandas, Matplotlib), R (Tidyverse)

Academic Workshops

Getting Started in Qualitative Education Research (ICOTS) <i>Co-facilitated with Kelly Findley and Nicola Sochacka. An gentle, 4hr introduction to qualitative research including study scoping, data collection methods and analysis, and use of theory. Structured workshop around a series of hands-on activities.</i>	2026
Dual-Benefit Student Reflection (Olin College) <i>Co-designed and ran a one-day workshop for Olin faculty on collaborative autoethnography as a means to promote and study student learning.</i> https://drive.google.com/drive/u/0/folders/1T25Fk06cBQBXXOo30gaRro7jFkaqJrm	2025
Data Science for Engineers, (Olin College) <i>Ran a 6-week professional development course for practicing engineers. Fifteen participants joined the course; five completed self-directed projects. Funded by NSF ERI grant.</i> https://zdelrosario.github.io/evc-course/frontmatter/home.html	2022 & 2023
Materials Informatics Workshop, (Olin College) Wellesley College <i>Adapted my MI Workshop materials for an undergraduate audience. Offered at Wellesley College in the Spring of 2022.</i> https://zdelrosario.github.io/mi101/intro.html	2022
Materials Informatics Workshop, (Olin College) Georgia Tech <i>Designing and facilitating an annual four-day workshop on materials informatics at Georgia Tech, sponsored by the Institute for Materials. Approximately 30 participants paid to attend the 2021 offering.</i> https://zdelrosario.github.io/mi101/intro.html	2021
Qualitative Data Workshop, (Stanford CTL) SABER WEST <i>Co-designed and facilitated a short workshop on qualitative data coding methods (verbal analysis) for ~ 30 participants.</i> https://sites.google.com/uci.edu/saberwest2020/program?authuser=0	2020
Materials Informatics Workshop, (Citrine Informatics) Georgia Tech <i>Designed and facilitated a two-day workshop on materials informatics at Georgia Tech, sponsored by the Institute for Materials. Led a team of 7 TA's to teach ~ 15 participants.</i> https://citrineinformatics.github.io/ga-tech-workshop/	2019
Teacher Workshops, (SeeME) Stanford <i>Developed and delivered workshops on the fundamentals of teaching, including lessons on learning goals and The 5E Model. Ran workshops for audiences of ~ 10.</i>	2019
Groupwork Workshop, (VPTL Consultant) Stanford <i>Co-developed and delivered workshop on evidence-based best-practices for groupwork in the classroom. Digested, summarized, and applied education literature, designed hands-on activities, co-facilitated workshop on several occasions for audiences of ~ 16.</i>	2018-2019

External Outreach

Teacher Development Workshop, (Individual) Olin <i>Interactive lesson on data visualization for the teaching staff at High Rock Middle School in Needham, MA.</i>	2022
Intro to Exploratory Data Analysis, (SeeME) Stanford <i>Introductory hands-on class to introduce students to principles of visualization, exploring data, understanding trends, and basic causal reasoning.</i> https://github.com/zdelrosario/teaching-eda	2016
"What the heck is engineering?", (Splash) Stanford <i>Introductory discussion-based class meant to introduce middle- and high-school aged students to engineering as a profession.</i>	2014-2015

Mentoring

PhD Defense Examiner	Institution, Department	Year
Gitanjali Bhattacharjee	Stanford University, Civil and Environmental Engineering	2021

Student Research Projects

- Alex George, Leslie Bostwick, Trinity Lee, Audrey Renaud, Emily Boyd, Sascha Jalinous, Liam Fries (Olin, 2023-Present) An ongoing collaborative autoethnography on engineering student learning. Resulted in peer-reviewed conference paper (George et al. 2025).
- Jin Ryu, Erika Saur, Jessica Lin (Wellesley, 2024-2025) Refinement of a survey-based instrument to measure data practices. Resulted in a journal submission (del Rosario, et al. 2025).
- Jin Ryu, Erika Saur (Wellesley, Su 2023) Prototyping of a survey-based instrument to measure data practices at scale. Resulted in a peer-reviewed conference paper (del Rosario, Ryu, and Saur 2024).
- Alex George, Leslie Bostwick, Trinity Lee (Olin, Su 2023) Initiation of a collaborative autoethnography on engineering students learning. Resulted in peer-reviewed conference paper (Bostwick et al. 2024).
- Ellen Sun, Maya Sivanandan (Olin, Su 2023) Design and implementation of statistical software.
- AJ Evans, Ellie Ramos, KD Vo, Emma Fox, Maeve Stites (Olin, Su 2022) Qualitative study of practicing engineers' data practices. Resulted in peer-reviewed journal paper (del Rosario 2024).
- Riya Aggarwal, Mira Flynn, Sam Daitzman (Olin, Su 2021) Qualitative study of engineering students about data practices. Published results in peer-reviewed conference paper (Aggarwal, et al. 2021).
- Emma Fox (Olin, Fa 2022) Qualitative coding scheme development and testing. Resulted in peer-reviewed conference paper (Fox and del Rosario, 2024).
- AJ Evans, KD Vo, Shreenithi Madan (Olin, Fa 2022) Scoping review of statistical content in engineering textbooks. Resulted in peer-reviewed conference paper (Vo et al. 2023).
- Riya Aggarwal, Caitlin Coffey, Arwen Sadler (Olin, 2020-2021) Thematic analysis of student experiences in online studio-based courses. Resulted in peer-reviewed conference paper (del Rosario, Aggarwal, et al. 2021).
- Qingmu (Josh) Deng (Olin, 2020-2021) Sequential design of experiments using gaussian processes. Co-advised with Rebecca Belisle (Wellesley). Resulted in conference presentation at Materials Research Society 2021.
- Gitanjali Bhattacharjee (Stanford, 2019) Sensitivity analysis for transit network reliability and bridge retrofitting. Resulted in peer-reviewed publication (Bhattacharjee and Baker 2021; *Structure and Infrastructure Engineering*).
- Sita Syal (Stanford, 2019) Soft cost analysis for solar farm leasing.
- Mark Benjamin (Stanford, 2018) Reliability-based design optimization strategies, focusing on density-matching.

Independent Studies Advised

- Dianna Sims (Olin, Sp 2023) Statistical process control and tolerance stackup analysis.
- Linda Hu (Olin, Sp 2023) Quantitative mechanical analysis.
- Shreenithi Madan (Olin, Sp 2022) Advanced data science: containerization, design of experiments, A/B testing, ANOVA.
- Shreenithi Madan (Olin, Fa 2022) SQL programming fundamentals.
- Colin Takeda (Olin, Fa 2022) Probability fundamentals.

- Jinfay (Justin) Yuan (Olin, Fa 2022) Data Science and design under uncertainty.

Writing Consultee Outcome

Paul Nadan	NASA NSTGRO; CMU, Robotics Institute	2020
Cindy Nguyen	NSF GRFP; Stanford, Electrical Engineering	2019
Larissa Little	NDSEG; Harvard, Materials Science	2018
Mason del Rosario	NSF GRFP Honorable Mention; UC Davis, Electrical Engineering	2018
Rongfei Lu	Stanford, Aeronautics and Astronautics	2018
Emma (Zeyan) Xu	NSF GRFP; Columbia, Material Science	2017

Service and Leadership

Getting Started in Statistics Education Research Webinar Series (CAUSE) <i>Co-organized a webinar series for new researchers in statistics education. Included sessions on scholarship vs. research, qualitative methods, and theoretical frameworks.</i>	2025 – Present
Academic Recommendations Board (Olin) <i>Served on a faculty judicial panel considering petitions related to Olin's curriculum. Designed and implemented a continuous improvement system to address an ABET accreditation deficiency. Served as chair starting in 2025.</i>	2024 – Present
Community Research Seminar Organizer (Olin College) <i>Organizing a seminar to highlight research conducted by the Olin community.</i>	2022 – 2024
TA Training Workshop (Olin College) <i>Co-designing and teaching a workshop on the fundamentals of teaching for TA's. Focused on active learning, problem solving, and the affective dimensions of teaching.</i>	2021 – 2024
Lead Instructor: ModSim (Olin College) <i>Organizing a teaching team of 2–4 faculty to teach a fall course on mathematical modeling for approximately 100 first-year students.</i>	2022 – Present
Selection Committee (AAC&U) <i>Aided in the selection the recipients of the 2022 Patricia Cross Future Leaders award, administered by the AAC&U. Conducted a read and rating of 12 candidate cases, performed a multi-criteria data analysis of all panelist ratings to aid in selection.</i>	Fall 2021
Racial Justice Discussion Organizer (Olin College) <i>Co-organized a 10-week series of discussions about racial justice at Olin College. Designed a session around BlackInTheIvory, highlighting Black voices in the academy. Facilitated difficult conversations about race in higher education.</i>	Summer 2020
Search Committee Member (Stanford Career Center, BEAM) <i>Served on executive hiring committee for Stanford's career center (BEAM). Interviewed BEAM staff to gain perspective; advocated for evidence-based structured interview process to reduce bias; co-wrote search criteria and rubric.</i>	2020
Co-Chair (ASEE, Task Force on Graduate Student Affairs) <i>Appointed by the president of the American Society for Engineering Education (ASEE) national organization to co-chair a task force studying how our professional society can serve graduate students.</i>	2019-2020
President (American Society for Engineering Education, Stanford Chapter) <i>Led and served on a 5-person organizing team. Organized a seminar sequence with internal and external speakers. Directed a Colloquium event attended by 70 persons, featuring workshops on "The Fundamentals of Teaching."</i>	2018-2019
Founder and Director of Curriculum (SeeME) <i>Co-founded Stanford Mechanical Engineering's student-run research outreach program SeeME. Developed and delivered workshops to train grad student instructors. Wrote grants to support operations and conference travel. Served as program leader and interfaced with the Department Chair.</i>	2017-2020
Financial Officer (American Society for Engineering Education, Stanford Chapter) <i>Served on 4-person organizing team. Wrote and won grants to fund speaker series and Colloquium. Point person on organizing seminar sequence.</i>	2017-2018
Chair of Teacher Development (Stanford Splash) <i>Served on 20-person organizing team serving thousands of high school students. Owned our teacher training program; co-facilitated workshops to introduce Stanford students to the basics of teaching. Enhanced teacher evaluations by introducing new survey system.</i>	2014-2016

Employment

Assistant Professor (Olin College)

September 2022-Present

Teaching first-year and upper-division courses on engineering and statistics. Conducting collaborative research on data science applied to materials development. Mentoring students in research projects.

Visiting Scientist (Toyota Research Institute)

January 2024-July 2024

Working as a data scientist on research projects to support Toyota's manufacturing quality.

Visiting Assistant Professor (Olin College)

September 2020-December 2021

Statistical Consultant (Citrine Informatics)

October 2019-December 2021

Supporting government-funded research projects with numerical and graphical statistical analyses. Co-author on resulting publications.

Instructional Designer and Data Scientist (Citrine Informatics)

Summer 2019

Developed and delivered 2-day workshop at Georgia Tech on Materials Informatics. Developed novel strategies for active learning in support of materials discovery.

Teaching Consultant (Stanford VPTL)

2018-2020

Professional teaching consultant, employed by the office of the Vice Provost for Teaching and Learning (VPTL). Used training in pedagogy and mentorship to carry out consultations with fellow graduate students. Co-facilitated various workshops, and co-developed novel workshop material on Groupwork.

Research intern (Northrop Grumman Corporation)

Summer 2017

Developed statistical methods to identify anomalies in time series data. Supported exoplanet detection research team.

Media Appearances

- [DrugDiscovery.net, August 2020](#)
- [ASEE Prism, October 2019](#)
- [Stanford Daily, January 2019](#)
- [Stanford News, April 2018](#)

The Analyst's Entreaty:
 "Grant me the insight to neglect the terms I do not need,
 Tenacity to understand strange interactions,
 And wisdom to know the (significant) difference."